Notes: Gumpert, Steimer & Antoni (QJE, 2022)
Firm Organization with Multiple Establishments

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1 Big picture

- **Question.** How does the geography of a multiestablishment firm shape its internal managerial hierarchy, and do shocks to one establishment propagate to headquarters and other establishments through managerial organization?
- **Core idea.** A multiestablishment firm shares a chief executive officer across headquarters and remote establishments. CEO time is scarce. Geographic frictions make it costly for remote workers to access CEO knowledge, so the firm reorganizes by hiring middle managers locally or at headquarters.
- **Main theoretical mechanism.** Middle managers substitute for CEO time. When a remote establishment becomes harder to help, hiring a layer at that establishment frees CEO time that can be reallocated to the headquarters and other establishments. The organization of one unit is therefore interdependent with the organization of other units.
- **Main empirical facts.** Distant establishments are less likely to exist and are smaller; multiestablishment firms with more distant establishments have more managerial layers; and firms usually adjust layers at headquarters or establishments separately rather than simultaneously.
- **Causal shock.** The paper uses openings of German high-speed railway routes, which exogenously reduce travel times between headquarters and establishments. The shock affects directly connected establishments, headquarters, and indirectly affected establishments in the same firm.
- **Main empirical finding.** Lower travel times increase employment at directly affected establishments and change organization at headquarters and other establishments. This supports the model's prediction that local changes propagate inside the firm through shared managerial resources.
- **Broader takeaway.** Local economic conditions do not remain local when establishments share managerial inputs. Managerial constraints are an internal network channel through which shocks travel across space.

2 Incremental contribution of the paper

- **From single-establishment hierarchies to multiestablishment hierarchies.** Prior hierarchy models largely explain the number of layers using firm size or knowledge-processing needs inside one establishment. This paper extends the hierarchy framework to firms with several locations and a shared CEO.
- **Interdependence of establishment organization.** The main incremental insight is that the optimal organization of one establishment depends on the characteristics and shocks of other establishments because they share scarce top-management time.
- **Managerial inputs are not pure public goods.** Multinational and multiunit firm models often treat headquarters inputs as nonrival public goods. The paper shows that managerial knowledge and CEO time can be rival and geographically costly to access.
- **New administrative measurement.** The paper uses German linked firm-establishment-employee data to measure managerial layers at headquarters and establishments. It observes all social-security employees, occupations, wages, establishments, and firms, then links them to Orbis firm data.
- **A unified theory and empirical test.** The paper develops a knowledge-hierarchy model and maps model objects directly to data: layers, managerial wage share, distance, travel-time frictions, establishment size, and wages.
- **Propagation through organization.** The paper adds managerial constraints as a propagation mechanism alongside financial constraints. A shock to one location affects organizational choices elsewhere in the same firm.
- **Causal evidence from infrastructure.** The high-speed-railway openings provide quasi-experimental variation in travel times. The paper uses the shock to show that both directly and indirectly affected units respond.

- **New lens for management heterogeneity.** The results help explain why management practices and layers differ across establishments within the same firm: each unit's organization depends on the firm's entire spatial portfolio.

3 Data and measurement

3.1 Administrative firm-establishment-employee data

- The data are linked German social-security, establishment, and firm records from 2000 to 2012.
- The employee data identify all employees subject to social security contributions on June 30 of each year, with occupation and wage.
- Firm balance-sheet and sales information comes from Orbis, linked to social-security data.
- The main panel uses 2000–2010 because 2011 has a change in occupational classification; cross-sectional fact tables use 2012.
- The main sample keeps firms with at least 10 employees and focuses on full-time employees.

Interpretation: The data are unusually strong because they observe both the firm and establishment hierarchy and allow the authors to distinguish headquarters from other establishments.

3.2 Definition of single- and multiestablishment firms

- A single-establishment firm consists only of headquarters.
- A multiestablishment firm consists of headquarters plus at least one other establishment.
- Headquarters is identified as the establishment located at the firm's postal code or locality.
- Establishments outside headquarters provide the spatial variation used to study geographic frictions.

Interpretation: The distinction matters because the model is about shared CEO time. A headquarters-only firm does not have the same internal spatial problem.

3.3 Managerial organization measures

- Employees are assigned to four occupational layers: production workers, supervisors/professionals, middle managers, and CEOs/managing directors.
- The preferred measure is the number of managerial layers above the lowest observed layer in the firm or establishment.
- Alternative measures are the share of wage sum earned by employees in managerial layers and by Blossfeld managerial occupations.
- These alternative measures are used to verify that results are not artifacts of one occupational classification.

Interpretation: The number of layers maps directly to the knowledge-hierarchy model, where layers represent a hierarchy of problem-solving knowledge.

3.4 Geographic frictions and high-speed railway treatment

- Cross-sectional geographic frictions are measured by distance from headquarters to establishments and by maximum distance to headquarters.
- The causal shock is lower rail travel time after openings of high-speed routes: Cologne–Frankfurt, Berlin–Hamburg, and Ingolstadt–Nuremberg.
- Treated establishments are those for which travel time to headquarters falls after route openings.
- Headquarters are treated if travel times to at least one establishment fall.
- Indirectly affected establishments are units in treated firms that do not directly experience a lower travel time themselves.

Interpretation: This definition lets the authors test both direct effects on the treated establishment and propagation effects within the firm.

4 Model objects and theoretical specifications

4.1 Knowledge production

One unit of labor generates a unit mass of problems. Knowledge is an interval of length z. If problems follow an exponential distribution, output from n units of labor and knowledge $\bar{z}$ is:

$$q = n\left(1 - e^{-\lambda \bar{z}}\right),$$

where λ measures predictability. A higher λ means a given knowledge stock solves more problems.

Interpretation: The model turns production into a problem-solving process. Layers exist because workers cannot solve every problem themselves and must ask more knowledgeable employees.

4.2 Learning and helping costs

Employees pay learning costs proportional to knowledge, and helping workers consumes time:

$$\text{learning compensation} = w(1 + cz), \qquad \text{helping cost} = \theta_{jk}.$$

Helping costs are higher across locations than within locations:

$$\theta_{EH} = \theta_{HE} \geq \theta_{HH} = \theta_{EE}.$$

Interpretation: Geographic frictions enter the model as a time cost of accessing headquarters knowledge. Distance or travel time makes the CEO more costly to use.

4.3 Layers and CEO time

- Production workers solve problems in their own knowledge interval.
- Middle managers solve problems that production workers cannot solve.
- The CEO is the highest-knowledge employee and is located at headquarters.
- The key assumption is that the firm has one CEO, whose time is a scarce resource shared across all units.

Interpretation: This scarce-CEO assumption is the paper's main theoretical lever. It makes each unit's organization depend on other units' demands for CEO help.

4.4 Organizational choice

For a given output vector, the firm chooses whether to operate in each location and how many layers to place below the CEO in headquarters and the establishment. The firm minimizes production costs subject to knowledge, time, and output constraints:

$$C(\tilde{q}) = \min_{\omega, \{n,z\}} \sum_{j \in \{H,E\}} \sum_{\ell} n^{\ell}_{j,\omega} w(1 + c z^{\ell}_{j,\omega}) + \text{CEO cost},$$

where ω summarizes organizational structure across locations.

Interpretation: The organization is a firm-level decision. Hiring a layer at one unit changes the demand for CEO time and therefore changes the optimal structure of other units.

4.5 Predictions

1. Higher geographic frictions make distant establishments less likely and smaller.
2. Higher geographic frictions raise the number of managerial layers at establishments and headquarters.
3. Firms usually add or drop a layer at either headquarters or establishments, but not both simultaneously.
4. Lower travel times reduce helping costs and should affect directly treated establishments, headquarters, and indirectly affected establishments through shared CEO time.

5 Empirical specifications and identification design

5.1 Cross-sectional distance regressions

For location probability, establishment size, and managerial layers, the paper estimates regressions of the form:

$$Y_{i,c} = \alpha + \beta \log Distance_{i,c} + X'_{i,c}\Gamma + FE + \varepsilon_{i,c}.$$

Key controls include market potential, relative wages, relative land prices, sector fixed effects, county fixed effects, legal-form fixed effects, and firm fixed effects when possible.

Interpretation: The cross-section documents the facts motivating the model. It does not by itself prove causality because firms choose where to locate establishments.

5.2 High-speed railway event regressions

The causal design uses openings of high-speed railway routes:

$$Y_{u,t} = \alpha_u + \delta_{c,t} + \beta Treated_{u,t} + \varepsilon_{u,t},$$

where u is an establishment or headquarters unit and $\delta_{c,t}$ represents county-year fixed effects. Treatment is a reduction in travel time to headquarters or from headquarters to at least one establishment.

Interpretation: The openings create plausibly exogenous reductions in geographic frictions, conditional on fixed effects and matched control groups.

5.3 Direct and indirect effects

The paper estimates effects separately for:

- directly affected establishments;
- headquarters of treated firms;
- indirectly affected establishments in treated firms.

Interpretation: This is the main test of interdependence. A direct local effect would affect only the treated establishment. The paper's theory predicts organizational responses elsewhere in the firm.

5.4 Matching and robustness

- Treated and control units are matched by size quartile, county, and year where possible.
- The paper uses weights from Iacus, King, and Porro coarsened exact matching.
- Robustness checks vary entry year, compare firms with one establishment, exclude indirectly affected establishments, and test event-time dynamics.

Interpretation: Robustness checks address strategic location, nonrandom route selection, pretrends, and the possibility that only direct establishments respond.

6 Section-by-section study guide

- **Introduction.** Read for the paper's thesis: local conditions propagate through internal organization because establishments share scarce CEO time.
- **Data.** Focus on how managerial layers are constructed from occupations and how headquarters/establishments are identified.
- **Facts.** Tables I–VI and Figures I–II establish the basic cross-sectional and dynamic facts about multi-establishment organizations.
- **Model.** Figures III–V explain average-cost functions, helping costs, and propagation mechanisms.
- **High-speed rail evidence.** Figure VI, Table VII, and Figures VII–VIII provide the quasi-experimental test.
- **Conclusion.** Focus on the broader implication: managerial constraints are a spatial propagation mechanism inside firms.

7 Tables and figures

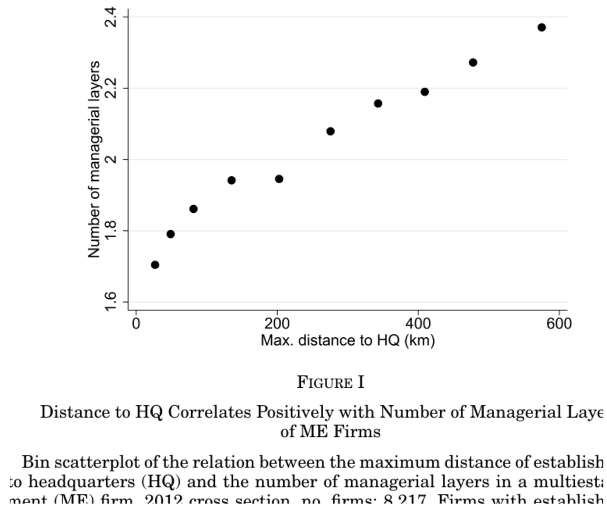
7.1 Figures I–II: Distance to headquarters and managerial layers

Read:

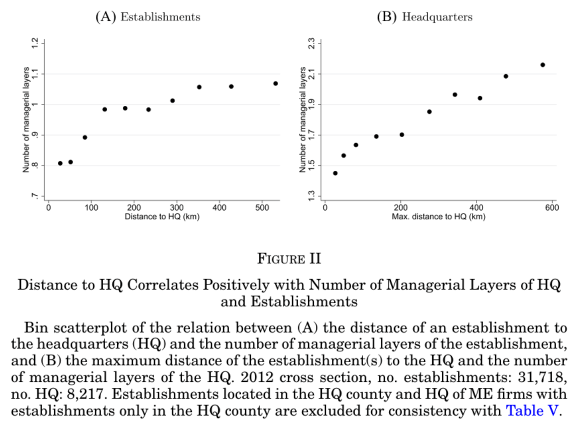
- Figure I shows that firms with establishments farther from headquarters have more managerial layers.
- Figure II decomposes this relation into establishment layers and headquarters layers.
- Both establishments and headquarters exhibit more layers as distance rises.

Detailed interpretation: The figures provide the first visual evidence for organizational interdependence. If only the remote establishment responded to distance, headquarters layers would not rise with maximum distance. The fact that headquarters layers rise indicates that distance strains the shared CEO resource.

Economic message: Distance is not merely a transportation variable. It is an organizational friction that changes how firms allocate managerial knowledge.



(a) Figure I: distance and layers in ME firms



(b) Figure II: establishment and HQ layers

7.2 Tables I–II: Descriptive statistics, SE versus ME firms and ME firms

Read:

- Table I shows that ME firms are only 9 percent of firms but account for 31.1 percent of establishments and 34.2 percent of employees.
- ME firms have more employees, larger sales, more managerial layers, and higher managerial shares than SE firms.
- Table II shows that the average ME firm has 4.6 establishments including headquarters and a mean maximum distance of 218 km to headquarters.
- Headquarters are much larger and have more layers than non-HQ establishments.

Detailed interpretation: These tables establish why multiestablishment firms matter. They are a small share of legal firms but a large share of employment. They are also structurally different: more layers, dispersed establishments, and concentrated management at headquarters.

Economic message: Large firms are not only larger versions of small firms. Their spatial organization creates a management-allocation problem.

TABLE I

DESCRIPTIVE STATISTICS, SE VERSUS ME FIRMS

Units of observation	N	of which ME firms (% share)							
Firms	109,357	9.0							
with nonmissing sales	57,811	9.2							
Establishments (incl. HQ)	144,437	31.1							
Employees	6,356,072	34.2							
Descriptive statistics	N	ME	Mean	Std. dev.	p25	p50	p75	p95	
No. employees	99,545	0	42	92	13	21	39	133	
	9,812	1	222	1,980	22	50	127	650	
Sales (M €)	52,524	0	28	694	2	4	9	67	
	5,287	1	358	4,111	4	15	74	608	
No. managerial layers	99,545	0	1.4	1.0	1	1	2	3	
	9,812	1	2.0	1.0	1	2	3	3	
Managerial share (%, layers)	99,545	0	28	28	5	19	43	88	
	9,812	1	36	29	11	29	58	90	
Managerial share (%, Blossfeld)	99,545	0	6	11	0	0	9	27	
	9,812	1	9	12	0	5	12	33	

Notes. 2012 cross section. *HQ*: headquarters; *ME*: multiestablishment firm; *No. employees*: number of full-time employees; *Sales (M €)*: sales in million €; *No. Managerial layers*: number of managerial layers; *Managerial share (%, layers/Blossfeld)*: share of wage sum earned by employees in managerial occupations (according to layers/Blossfeld occupational categories).

Cross-checking the results regarding the number of layers with the managerial share ensures that our results are robust to this possibility. We determine managerial occupations in two ways.

(a) Table I: SE versus ME descriptive statistics

TABLE II

DESCRIPTIVE STATISTICS, ME FIRMS

Descriptive statistics, firm	N		Mean	Std. dev.	p50	p75	p95
No. establishments (incl. HQ)	9,812		4.6	19.6	2	3	10
Maximum distance to HQ, km	9,812		218	189	167	376	547
Minimum area covered, km^2	3,579		30,117	41,725	7,025	49,915	125,253
Descriptive statistics, HQ/ establishment	N	HQ	Mean	Std. dev.	p25	p50	p75
No. employees	35,080	0	32	333	2	5	16
	9,812	1	107	669	11	27	76
No. managerial layers	35,080	0	1.0	0.8	0	1	2
	9,812	1	1.7	1.1	1	2	3
Managerial share (%, layers)	35,080	0	37	38	0	24	70
	9,812	1	38	32	11	32	64
Managerial share (%, Blossfeld)	35,080	0	8	19	0	0	5
	9,812	1	10	16	0	4	14

Notes. 2012 cross section. *No. establishments (incl. HQ)*: number of establishments (including headquarters); *Maximum distance to HQ, km*: maximum distance between establishments and HQ in kilometers, where distance is computed as the population-weighted average of the distances between all municipalities in the establishment county and the HQ county; *Minimum area covered, km^2*: minimum area covered by establishments and HQ in square kilometers, only available for firms with at least two establishments in addition to HQ.

firms. They make up a disproportionate share of establishments

(b) Table II: ME descriptive statistics

7.3 Tables III–IV: Cross-sectional distance facts

Read:

- Table III shows that distance to headquarters is negatively related to establishment location probability and establishment size.
- Table IV shows that maximum distance to headquarters is positively related to firm-level managerial layers and managerial shares.
- The results survive controls for market potential, wages, land prices, sector/county/legal-form fixed effects, and firm fixed effects where relevant.

Detailed interpretation: Together, the tables describe two margins of geography: extensive/intensive location choices and organizational complexity. Distance discourages establishment presence and size, but conditional on operating remotely, firms adopt more layers.

Economic message: Geographic frictions operate through both where firms operate and how they manage the operations they keep.

TABLE III

DISTANCE TO HQ CORRELATES NEGATIVELY WITH LOCATION PROBABILITY AND ESTABLISHMENT SIZE

	Location probability			Log no. est. employees		
	(1)	(2)	(3)	(4)	(5)	(6)
Log distance to HQ	− 0.315***	− 0.303***	− 0.368***	− 0.106***	− 0.112***	− 0.137
	(0.021)	(0.023)	(0.020)	(0.018)	(0.019)	(0.017
Log market potential	0.745***	0.780***		0.485***	0.465***	
	(0.026)	(0.031)		(0.044)	(0.046)	
Relative wages	− 0.942***	− 0.887***		− 0.330**	− 0.433***	
	(0.062)	(0.063)		(0.108)	(0.109)	
Relative land prices		− 0.021***			0.020***	
		(0.005)			(0.005)	
No. observations	3,715,666	3,222,108	3,715,666	21,496	19,203	21,49
No. firms	9,266	8,732	9,266	3,006	2,773	3,00
HQ sector FE	Y	Y	Y	N	N	N
HQ county FE	Y	Y	Y	N	N	N
Legal form FE	Y	Y	Y	N	N	N
County FE	N	N	Y	N	N	Y
Firm FE	N	N	N	Y	Y	Y
Model		Probit			OLS	

Notes. 2012 cross section. The table presents the coefficient estimates of a probit model in columns (1)–(3) (a constant is included; standard errors clustered by HQ county in parentheses) and a linear model in columns (4)–(6) (standard errors clustered by firm and county are in parentheses). The regressions in columns (4)–(6) control for firm f effects, hence they only include ME firms with establishments in at least two counties. ** $p < .01$, *** $p < .001$. Dependent variable: columns (1)–(3): indicator for whether fi owns at least one establishment in county c; columns (4)–(6): log number of employees of establishment(s) in county c. Independent variables: *Log distance to HQ*: log dista between county c and HQ county of firm i in km; *Log market potential*: log of average of GDP of county c and surrounding counties weighted by distance; *Relative wages/land pr* average wages/land prices in county c relative to wages/land prices in HQ county of firm i. We compute average wages in a county excluding firm i. Distance, market potential, relative land prices are computed using data of the German Federal Statistical Office. The number of firms is lower than the number of ME firms due to missing values for legal form. FE = fixed effects.

(a) Table III: distance, location probability, and size

TABLE IV

DISTANCE TO HQ CORRELATES POSITIVELY WITH THE NUMBER OF MANAGERIAL LAYERS OF ME FIRMS

	No. managerial layers				Mg. share ∈ [0, 1]		Mg. share ∈ [0,	
					Layers		Blossfeld	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(
Maximum log distance to HQ	0.039***		0.050***		0.152***		0.131***	
	(0.007)		(0.006)		(0.014)		(0.019)	
Log area		0.025***		0.027***		0.070***		0.0
		(0.005)		(0.004)		(0.011)		(0.0
Log sales	0.115***	0.082***						
	(0.004)	(0.006)						
Log no. non-mg. employees			0.109***	0.088***				
			(0.004)	(0.005)				
No. firms	4,323	1,661	7,742	2,768	7,742	2,768	7,742	2,
HQ sector FE	Y	Y	Y	Y	Y	Y	Y	
HQ county FE	Y	Y	Y	Y	Y	Y	Y	
Legal form FE	Y	Y	Y	Y	Y	Y	Y	
Model		Poisson					GLM	

Notes. 2012 cross section. The table presents the coefficient estimates. A constant is included. Robust standard errors are in parentheses. *** $p < .001$. Even columns inclu ME firms active in at least three counties. The generalized linear model (GLM) assumes a logit link function and the binomial distributional family. Dependent variable: co (1)–(4) number of managerial layers, columns (5)–(6) managerial share in wage sum, according to layers, columns (7)–(8) managerial share in wage sum, according to Bl occupational categories. Independent variables: *Maximum log distance to HQ*: log of maximum distance between establishment and headquarters in km; *Log area* spanned b log of minimum area covered by establishments and HQ in square kilometers; *Log sales*: log annual sales; *Log no. non-mg. employees*: log number of employees at lowest lay = fixed effects, mg. = managerial.

(b) Table IV: distance and ME layers

7.4 Tables V–VI: Unit-specific layers and transition dynamics

Read:

- Table V shows that distance to headquarters is positively associated with layers and managerial shares both at establishments and headquarters.
- Table VI shows that firms tend to change layers at headquarters or establishments, but rarely at both simultaneously.
- The transition matrix reveals high persistence, with most firms keeping the same number of layers.

Detailed interpretation: Table V is a direct cross-sectional test of the model's key allocation logic. Table VI adds dynamic evidence: layers adjust in discrete and unit-specific ways, consistent with middle managers being a quasi-fixed organizational investment.

Economic message: Multiunit organizational change is lumpy and interdependent. Firms usually reallocate managerial structure at one part of the hierarchy rather than changing all units at once.

TABLE V

DISTANCE TO HQ CORRELATES POSITIVELY WITH THE NUMBER OF MANAGERIAL LAYERS OF THE HQ AND ESTABLISHMENTS

	Establishment			Headquarters		
	No. layers	Mg. share $\in [0, 1]$		No. layers	Mg. share $\in [0, 1]$	
		Layers	Blossfeld		Layers	Blos
	(1)	(2)	(3)	(4)	(5)	(
Log distance to HQ	0.056***	0.191***	0.236***			
	(0.011)	(0.032)	(0.053)			
Maximum log distance to HQ				0.057***	0.166***	0.1
				(0.007)	(0.016)	(0.0
Log no. non-mg. employees	0.256***			0.162***		
	(0.010)			(0.004)		
No. est./HQ	26,409	31,717	31,717	7,999	8,217	8,
Sector FE	Y	Y	Y	Y	Y	
County FE	Y	Y	Y	Y	Y	
Model	Poisson	GLM		Poisson	GLM	

Notes. 2012 cross section. The table presents the coefficient estimates. A constant is included. Standard errors are in parentheses (clustered by firm in columns (1)–(3), in columns (4)–(6)). *** $p < .001$. The generalized linear model (GLM) assumes a logit link function and the binomial distributional family. Dependent variable: columns ((4) number of managerial layers, columns (2) and (5) managerial share in wage sum, according to layers, columns (3) and (6) managerial share in wage sum, according to Bl occupational categories. Independent variables: *Log distance to HQ*: log of distance between establishment and headquarters in km; *Maximum log distance to HQ*: log of ma distance between establishment and HQ in km; *Log no. of non-mg. employees*: log number of employees at lowest layer in establishment/HQ. FE = fixed effects, mg. = mana

(a) Table V: distance and layers at HQ and establishments

TABLE VI

TRANSITION DYNAMICS OF THE MANAGERIAL ORGANIZATION

Panel A: No. managerial layers of firm

No. layers in $t/t+1$	0	1	2	3	SE	No. fi
0	**85**	8	1		6	10,7
1	5	**81**	8	1	6	18,2
2		7	**79**	8	5	18,7
3			6	**90**	4	22,3

Panel B: No. managerial layers at headquarters (HQ)/establishments (est.)

No. layers in $t/t+1$	0/0	1/0	1/1	2/<2	2/2	3/<3	3/3	SE	No. fi
HQ 0/ est. 0	**85**	5						6	10,7
HQ 1/ est. 0	6	**75**	4	6				8	8,3
HQ 1/ est. 1	1	5	**76**	7		1		4	8,0
HQ 2/ est. 0,1		4	4	**76**	2	6		7	12,0
HQ 2/ est. 2			1	10	**69**	9	1	2	3,4
HQ 3/ est. 0,1,2				5	2	**84**	3	5	13,3
HQ 3/ est. 3						9	**86**	1	4,6

Notes. 2000–2010 data. Panel A displays the percentage share of firms that transition from a number of managerial layers in year t (given in the rows) to a potentially diff number of layers or to SE firm status in year $t+1$ (given in the columns). Panel B displays the percentage share of firms that transition from a managerial organization in given in the rows) to a potentially different managerial organization or to SE firm status in year $t+1$ (given in the columns). The figure in front of the slash denotes the nu of layers of the HQ. The figure behind the slash denotes the maximum number of layers of the establishments. Firms with a higher number of layers at the establishments at the HQ are dropped for readability. Empty cells contain fewer than 0.5% of firms. Fewer than 0.5% of firms exit. The diagonal is in bold.

(b) Table VI: transition dynamics

7.5 Figures III–IV: Average-cost functions and transport frictions

Read:

- Figure III compares average-cost functions for SE and ME firms without transport frictions.
- Figure IV shows how transport and helping-cost frictions alter the output thresholds at which the firm reorganizes.
- In ME firms, several organizational structures may be relevant: no middle managers, a layer at one unit, or layers at both units.

Detailed interpretation: These figures translate the hierarchy model into cost curves. Adding a layer is costly at small scale but reduces problem-solving costs at larger scale. Transport frictions shift the thresholds and can make local middle managers desirable earlier.

Economic message: The number of layers is a cost-minimizing organizational response to the scale of operations and the difficulty of accessing higher-level knowledge.

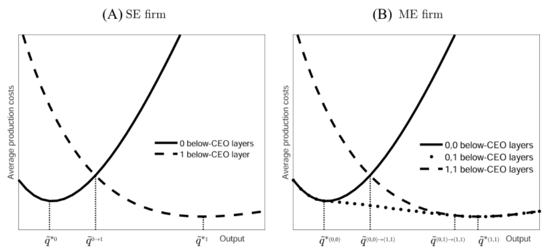


FIGURE III

Illustration of the Average-Cost Function, no Transport Frictions

The figure plots the average-cost functions of a SE and a ME firm for $\tau = 1$, $\theta_{HH} = \theta_{EH}$. Parameter values: $\frac{c}{\lambda} = 0.225$, $\theta_{HH} = 0.26$ (from CRH), $w = 1$. Panel A: The average-cost function of a SE firm is U-shaped given L. The firm adds a layer at $\tilde{q}^{0\to 1}$. Panel B: The average-cost function of a ME firm with a symmetric number of below-CEO layers is U-shaped. The firm adds a layer at the establishment at $\tilde{a}^{*(0,0)}$ and at the headquarters at $\tilde{a}^{(0,1)\to(1,1)} > \tilde{a}^{(0,0)\to(1,1)}$ (or vice

(a) Figure III: average costs without transport frictions

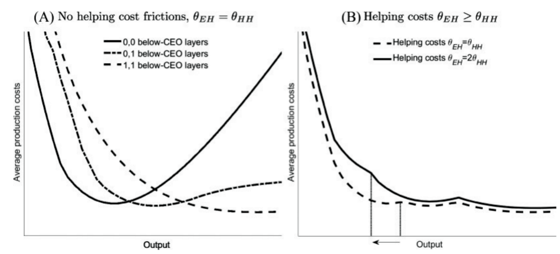


FIGURE IV

Illustration of the Average-Cost Functions, Transport Frictions

The figure plots average-cost functions of a ME firm. Parameter values: $\frac{c}{\lambda} = 0.225$, $\theta_{HH} = 0.26$ (from CRH), $w = 1$, $\tau = 1.1$, $\tilde{q}_H = \tilde{q}_E$. Panel A: at each kink, the ME firm adds a layer at one unit. The (0,1) and (1,0)-organization have the same costs. Panel B: higher helping costs θ_{EH} decrease the output at which

(b) Figure IV: average costs with transport frictions

7.6 Figures V–VI: Model predictions and high-speed railway routes

Read:

- Figure V summarizes the causal chain from travel times to helping costs, output allocation, CEO knowledge/time allocation, and organizational structure.
- Figure VI maps the new high-speed railway routes and reports the travel-time reductions: Cologne–Frankfurt, Berlin–Hamburg, and Ingolstadt–Nuremberg.
- The routes reduce travel times by large amounts and affect cities beyond the route endpoints because the German rail network is highly connected.

Detailed interpretation: Figure V is the conceptual bridge from the model to the empirical design. Figure VI is the empirical shock. The model says lower travel time lowers the cost of helping remote workers; the map shows where such shocks occur.

Economic message: Infrastructure affects firm organization because it changes the cost of transferring managerial knowledge across space.

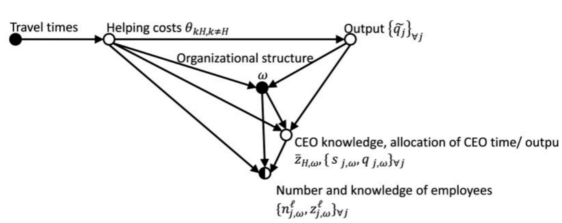
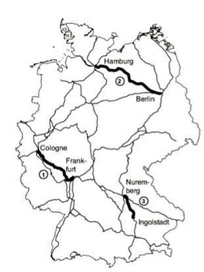


FIGURE V

Model Predictions Regarding the Effects of a Change in the Travel Times

The graph illustrates the predictions of the model regarding the effects of change in travel times. The arrows denote causal relationships between the var

(a) Figure V: model predictions



Route	Date of opening	Travel time before	Travel time after	Change
1) Cologne—Frankfurt	Aug. 2002	135 min	76 min	-44%
2) Berlin—Hamburg	Dec. 2004	135 min	90 min	-33%
3) Ingolstadt—Nuremberg	May 2006	66 min	30 min	-55%

FIGURE VI

The New High-Speed Railway Routes and the German Long-Distance Network

The map shows the German long-distance rail network (light) and the new high-speed railway routes (bold). Data from Deutsche Bahn AG (http://data.deutschebahn.com/dataset/geo-strecke).

(b) Figure VI: high-speed railway routes

7.7 Table VII: Lower travel times affect all units of ME firms

Read:

- Directly affected establishments grow: the log number of nonmanagerial employees rises by about 0.084 in the all-firm sample.
- For headquarters of firms with at least two establishments, wages, layers, and managerial share rise after travel times fall.
- Indirectly affected establishments show increases in wages and managerial occupation share.
- The table uses fixed effects and matched control units.

Detailed interpretation: This is the main quasi-experimental table. The direct employment effect is expected if travel-time reductions make remote operations more viable. The headquarters and indirect-establishment responses are more distinctive: they support the model's prediction that managerial reallocation propagates across the firm.

Economic message: Local infrastructure shocks reorganize firms internally. The effect is not contained at the treated establishment.

TABLE VII

LOWER TRAVEL TIMES AFFECT ALL UNITS OF ME FIRMS

	All firms				Firms with $\geqslant$ 2 establishments			
	No. em. (1)	Wages (2)	No. lay. (3)	Mg.sh. (4)	No. em. (5)	Wages (6)	No. lay. (7)	Mg.sh. (8)
Directly affected establishment								
Est. treated	0.084***	0.002	0.009	0.059	0.083***	0.004	0.016	0.049
	(0.019)	(0.004)	(0.016)	(0.253)	(0.021)	(0.005)	(0.018)	(0.273)
No. observations	47,732	47,732	47,732	47,732	40,143	40,143	40,143	40,143
No. est.	5,609	5,609	5,609	5,609	4,791	4,791	4,791	4,791
R-squared	0.901	0.931	0.875	0.890	0.899	0.932	0.878	0.902
Est. FE	Y	Y	Y	Y	Y	Y	Y	Y
County-year FE	Y	Y	Y	Y	Y	Y	Y	Y
Headquarters								
Firm treated	– 0.013	0.004	0.019	0.103	0.023	0.018*	0.065**	0.994+
	(0.019)	(0.005)	(0.022)	(0.338)	(0.033)	(0.008)	(0.022)	(0.504)
No. observations	13,393	13,393	13,393	13,393	6,261	6,261	6,261	6,261
No. HQ	1,469	1,469	1,469	1,469	683	683	683	683
R-squared	0.951	0.951	0.875	0.913	0.951	0.945	0.872	0.919
HQ FE	Y	Y	Y	Y	Y	Y	Y	Y
HQ c.-year FE	Y	Y	Y	Y	Y	Y	Y	Y
Indirectly affected establishment								
Firm treated					– 0.017	0.019***	0.020	0.521*
					(0.020)	(0.004)	(0.017)	(0.209)
No. observations					45,508	45,508	45,508	45,508
No. est.					5,508	5,508	5,508	5,508
R-squared					0.917	0.926	0.890	0.887
Est. FE					Y	Y	Y	Y
County-year FE					Y	Y	Y	Y
HQ c.-year FE					Y	Y	Y	Y

Notes. 2000–2010 panel. Standard errors clustered by (headquarter) county are in parentheses. $^{+}$ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$. Dependent variables: *No. em.*: log number of nonmanagerial employees; *Wages*: average log wages of nonmanagerial employees; *No. lay.*: number of managerial layers; *Mg.sh.*: share of managerial occupations in wage sum in percent, where managerial occupations are determined according to Blossfeld occupational categories. Treated and control units are matched by size quartile, (headquarters) county, and year, except for the indirectly affected establishments that are matched by size quartile and year. We use the weights of Iacus, King, and Porro (2012) to estimate the average treatment effect on the treated. All variables are winsorized at the first and 99th percentiles. The *p*-value for the managerial share of the headquarters in column (8) is 5.4%.

Table VII: lower travel times affect all units of ME firms

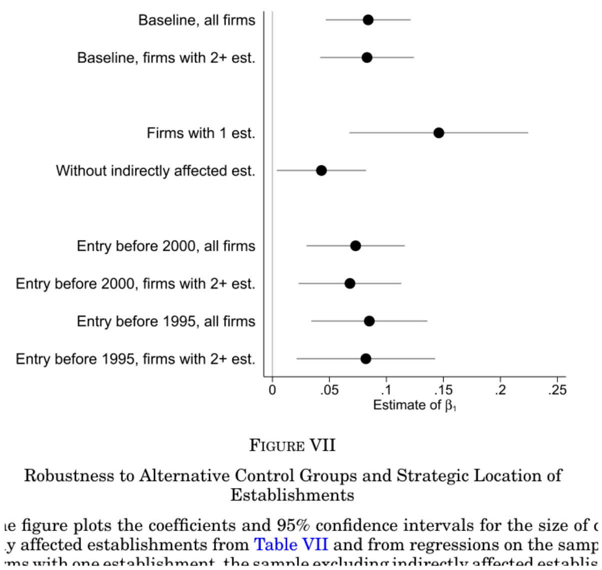
7.8 Figures VII–VIII: Robustness and event-time evidence

Read:

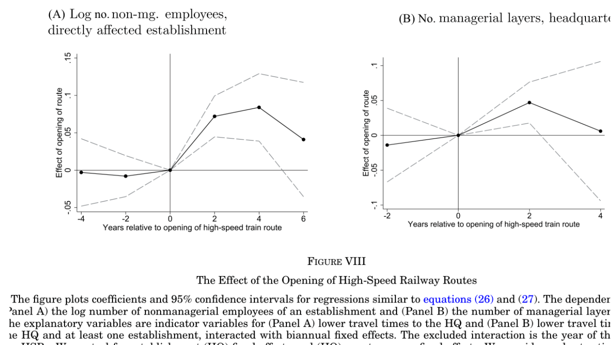
- Figure VII shows that the direct establishment employment effect is robust across alternative control groups and sample restrictions.
- Figure VIII plots event-time coefficients for directly affected establishments and headquarters.
- The event-time pattern supports the interpretation that changes occur after high-speed-rail openings rather than before.

Detailed interpretation: Figure VII addresses concerns about strategic location choices and alternative counterfactuals. Figure VIII addresses pretrend concerns. Together, they make the high-speed-rail evidence more credible.

Economic message: The causal evidence is not driven by one matching choice or by obvious pretrends; it reflects real organizational adjustment to lower geographic frictions.



(a) Figure VII: robustness to alternative controls



(b) Figure VIII: event-time effects

8 Result map

- **Fact 1.** Distance lowers the probability that a firm operates an establishment and lowers establishment size.
- **Fact 2.** Distance raises managerial layers at both establishments and headquarters.
- **Fact 3.** Firms reorganize by changing layers either at headquarters or at establishments, not both simultaneously in most cases.
- **Theory.** Geographic frictions raise the cost of accessing CEO knowledge, and middle managers substitute for CEO time.
- **Causal evidence.** High-speed rail reduces travel times and triggers changes in directly affected establishments, headquarters, and indirectly affected establishments.
- **Mechanism.** The effects are consistent with changes in helping costs and CEO-time allocation, not simply local demand or output changes.

9 Short summary

- The paper studies managerial organization in multiestablishment firms.
- It uses administrative German firm-establishment-employee data.
- Multiestablishment firms account for a large share of employees and have more managerial layers than single-establishment firms.

- Establishments are less likely and smaller when farther from headquarters.
- More distance is associated with more managerial layers at establishments and headquarters.
- The model treats CEO time as scarce and shared across firm units.
- Middle managers substitute for CEO help and release CEO time.
- High-speed railway openings reduce travel times and provide quasi-experimental evidence.
- Lower travel times affect direct establishments, headquarters, and indirectly affected establishments.
- The main lesson is that local shocks propagate through firm organization because managerial inputs are rival and spatially costly to access.

10 Conclusion

10.1 Core conclusion

Gumpert, Steimer, and Antoni show that the internal organization of multiestablishment firms is interdependent across establishments. The number of managerial layers at one unit depends on conditions at other units because all units share scarce CEO time.

10.2 Economic intuition

The CEO is the firm's top knowledge resource. When a remote establishment is costly to help, the firm can either spend more CEO time or hire local middle managers. Hiring middle managers at one unit changes how much CEO time is available elsewhere. This creates cross-establishment organizational spillovers.

10.3 Incremental contribution

The paper extends hierarchy theory beyond single-location firms, measures layers inside German firms, and provides causal evidence that reducing geographic frictions reorganizes multiple units of a firm. The contribution is both theoretical and empirical: it models the rival nature of managerial inputs and then tests the mechanism with high-speed railway shocks.

10.4 Implications

The results imply that local infrastructure, local shocks, and local conditions can have nonlocal effects through organizational design. They also warn against treating headquarters knowledge as a pure public good within firms. Managerial attention and CEO time are scarce resources.

10.5 Limitations

The high-speed railway design is credible but not a randomized experiment. The paper cannot observe CEO time directly, and the model abstracts from monitoring hierarchies, information technology, and many internal processes. Still, the evidence strongly supports a managerial-constraint channel.

10.6 Takeaway

Multiestablishment firms are internal networks. Understanding their organization requires studying the whole firm rather than isolated establishments.